Penchala Siva Prasanth Kasa

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Available for full-time employment from June 2027 (M.S. completes May 2027).

Summary

AI/ML engineer with production experience across model inference, retrieval and deployment, now an M.S. in CS at UW–Madison. At Qualcomm, used **Python** to benchmark, profile and tune a heterogeneous inference runtime serving **LLM** and vision models, informing cost-aware operator routing. Built a distributed **vector search** engine over embeddings scored on recall@k, and shipped ML scoring to production at sub-10ms over millions of daily transactions. Dual degree from IIT Kharagpur; **JEE Mains Paper-II All-India Rank 23**.

Experience

AI Software Intern @ Qualcomm, Inc., *San Diego, USA* May 2026 - Aug. 2026

- Wrote **Python** benchmarking and **model evaluation** harnesses that scored every change on latency before it shipped.
- Profiled and tuned cross-device data movement and sync in **Python**, informing cost-aware operator routing decisions.
- Contributed C++ to Qualcomm's ExecuTorch backend for graph partitioning of a runtime serving **LLM** and vision models.

Software Engineer @ Enphase Energy, *Bangalore, India | 2 years, full-time* Jul. 2023 - Jul. 2025

- Shipped production **C/C++** services for a metering device, bootloader through application, on Mbed OS/Zephyr.
- Cut RAM 98% to 82% and storage by 20% by profiling data paths and redesigning memory layouts and data structures.
- Hardened multithreaded services and Protobuf encoding for real-time telemetry over CAN/Zigbee; triaged field incidents.
- Scripted build, flash and regression checks in Bash/Python behind CI; drove design docs and code reviews in sprints.

Software Intern @ Visa, Inc., *Bangalore, India* May 2022 - Jul. 2022

- Shipped **ML models** to production: **sub-10ms** scoring over **millions of transactions daily**, with guardrails.
- Automated **model onboarding** with Go/Java microservices over Kafka and Spark, saving 2 weeks of work per model.

AI & Machine Learning Projects

VecScale - Distributed Vector Search over Embeddings (C++) | *UW-Madison* Jan. 2026 - May 2026

- Built a distributed **vector search** engine over an embedding corpus: sharded index, parallel top-k workers.
- Scored retrieval quality with **recall@k** against an exact baseline; MPI, OpenMP, CUDA and Slurm for benchmarking.

Whisper-to-Speech Conversion | Dual Degree Thesis, Dept. of AI, *IIT Kharagpur* Aug. 2022 - May 2023

- Trained an end-to-end **deep learning** speech model (autoencoder + **Bi-LSTM**, **PyTorch**/Keras) on audio time series.
- Reached a **12.24% word error rate** through hypothesis-driven ablations; presented at Empower 2025.

AURA - On-Device Multimodal Assistant | Qualcomm Hackathon, *UW-Madison* Feb. 2026

- Paired vision-language models (CLIP/BLIP) with speech-to-text (Whisper) for real-time on-device scene understanding.
- Served multi-model inference locally on **ONNX Runtime** and the Qualcomm QNN backend with CPU fallback.

Job Board Crawler & Change Detector (Python) | *Personal* 2025 - 2026

- Built a crawler that scans **thousands of postings across 62 job boards** nightly and emails only the new matches.

Technical Skills

- **Languages:** Python, C++, C, SQL, Go, Java, Bash, JavaScript/TypeScript
- **Machine Learning:** PyTorch, TensorFlow, ONNX Runtime, ExecuTorch, scikit-learn, model evaluation, ablations
- **AI & Retrieval:** LLM serving & inference, vector search, embeddings, recall@k, prompt engineering, AI agents (coursework)
- **Cloud & Systems:** AWS, Azure, Docker, CI/CD, REST APIs, microservices, Kafka, Spark, Linux
- **Engineering:** Data structures & algorithms, system design, concurrency, Git, gdb, code review, Agile

Education

University of Wisconsin, Madison, USA Sep. 2025 - May 2027
Master of Science (M.S.) in Computer Science *School of Computer, Data & Information Sciences*

- **Coursework:** Machine Learning, AI Agents, Computer Vision, High Performance Computing, Databases.
- **Graduate Teaching Assistant** for CS 640: Computer Networks.

Indian Institute of Technology, Kharagpur, India July. 2018 - May 2023
Dual Degree - B.Tech(Hons.) + M.Tech (5-year integrated) *Electronics & Electrical Communication Engineering*

- **Coursework:** ML/AI, Digital Image Processing, Data Structures & Algorithms.

Achievements

- **Nationally Ranked 23** among 125,000 candidates for IIT JEE Mains Paper-II (2018). (Top 0.018%)
- Nationally Ranked 1318 among 230,000 candidates for IIT JEE Advanced (2018). (Top 0.5%)
- Nationally Ranked 1250 in the Graduate Aptitude Test in Engineering (GATE - 2022). (Top 1%)